

ZHIGANSK, Russian Federation

WMO: 243430

Lat: **66.767N**Lon: **123.400E**Elev: **302**StdP: **14.54**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-55.0	-52.3	-58.3	0.2	-55.0	-55.5	0.2	-52.0	16.8	-16.0	14.4	-20.4	2.6	140	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	80.1	63.5	76.4	62.4	72.8	60.7	66.6	74.9	64.4	72.8	62.3	71.1	6.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	88.8	70.9	61.1	81.2	68.7	58.5	74.0	66.6	31.5	74.5	29.8	72.9	28.2	71.2	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.7	13.1	DB	-58.2	86.1	4.6	3.3	-61.6	88.5	-64.2	90.4	-66.8	92.3	-70.2	94.8	(4)
(5)			WB	-58.8	70.1	4.8	2.6	-62.3	72.0	-65.1	73.5	-67.8	74.9	-71.3	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	-34.4	-28.9	-7.1	16.7	36.4	55.7	61.1	53.9	38.8	14.6	-17.9	-34.6	(6)	
	DBStd	36.55	11.18	11.40	13.66	11.10	9.34	8.75	7.57	6.75	7.61	13.33	12.81	11.71	(7)	
	HDD50	14207	2617	2210	1771	998	429	42	5	36	343	1097	2038	2621	(8)	
	HDD65	19015	3082	2630	2236	1448	885	296	169	348	785	1562	2488	3086	(9)	
	CDD50	737	0	0	0	0	8	214	349	158	8	0	0	0	(10)	
	CDD65	70	0	0	0	0	0	17	48	5	0	0	0	0	(11)	
	CDH74	582	0	0	0	0	0	168	382	32	0	0	0	0	(12)	
	CDH80	97	0	0	0	0	0	28	66	3	0	0	0	0	(13)	
(14) Wind	WSAvg	5.8	5.2	4.9	5.6	6.7	7.3	6.8	6.6	5.9	5.9	5.8	4.7	4.6	(14)	
(15) Precipitation	PrecAvg	10.8	0.4	0.3	0.4	0.5	0.7	1.3	1.9	1.5	1.4	1.2	0.7	0.5	(15)	
	PrecMax	14.3	0.6	0.9	1.2	1.1	2.2	3.3	6.3	3.4	2.3	1.9	1.7	1.1	(16)	
	PrecMin	7.2	0.1	0.1	0.1	0.0	0.0	0.2	0.0	0.4	0.6	0.5	0.2	0.2	(17)	
	PrecStd	2.1	0.2	0.2	0.3	0.3	0.6	0.8	1.5	0.8	0.5	0.3	0.4	0.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.5	5.3	30.7	47.2	68.6	83.8	85.5	77.6	66.4	37.7	17.3	-1.0	(19)	
		MCWB	-3.0	4.9	26.7	37.5	51.5	63.7	65.5	63.9	53.2	35.6	16.3	-1.4	(20)	
	2%	DB	-8.1	-1.7	23.9	41.1	59.8	78.7	81.6	72.8	59.6	34.7	11.6	-8.3	(21)	
		MCWB	-8.5	-2.2	21.2	33.6	46.4	60.9	64.7	61.7	50.0	32.9	11.1	-8.8	(22)	
	5%	DB	-13.2	-7.0	18.7	37.4	55.3	74.7	78.5	69.6	54.3	32.4	6.6	-12.4	(23)	
		MCWB	-13.6	-7.4	16.7	31.9	44.0	59.4	63.8	59.7	47.7	31.1	6.0	-12.8	(24)	
	10%	DB	-18.2	-11.9	13.6	34.1	50.9	71.0	75.0	66.1	50.3	30.3	0.4	-16.9	(25)	
		MCWB	-18.5	-12.3	11.9	29.8	41.5	57.6	63.0	57.9	45.3	28.8	-0.2	-17.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.0	4.8	27.2	38.2	51.4	66.5	70.4	67.7	55.2	36.4	16.4	-1.5	(27)	
		MCDB	-2.6	5.3	30.5	45.7	67.4	78.6	79.9	75.5	63.1	37.2	17.2	-1.0	(28)	
	2%	WB	-8.4	-2.3	21.5	34.8	47.5	63.6	67.8	63.9	51.8	33.3	11.1	-8.7	(29)	
		MCDB	-8.1	-1.9	23.7	39.8	57.6	74.3	75.9	69.4	56.0	34.1	11.5	-8.4	(30)	
	5%	WB	-13.5	-7.4	16.8	32.4	44.8	61.5	65.9	61.3	48.5	31.5	6.0	-12.7	(31)	
		MCDB	-13.1	-7.0	18.6	36.8	53.8	72.1	74.1	67.6	53.5	32.4	6.5	-12.4	(32)	
	10%	WB	-18.5	-12.2	11.8	29.7	42.3	58.9	64.0	59.0	45.6	28.9	-0.1	-17.2	(33)	
		MCDB	-18.2	-11.9	13.4	34.4	50.4	68.5	72.7	64.6	49.9	30.2	0.3	-16.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.0	16.5	18.5	14.1	16.9	16.3	15.7	12.8	9.8	10.6	9.1	(35)	
		MCDBR	14.1	13.7	17.4	18.1	19.4	23.2	22.2	21.1	20.7	8.0	12.5	12.5	(36)	
	5% WB	MCWBR	14.0	13.5	15.6	13.0	10.9	11.1	10.2	11.7	11.9	7.3	12.3	12.4	(37)	
		MCWBR	14.1	13.6	17.3	16.4	17.9	20.9	17.4	17.6	18.4	7.8	12.5	12.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.186	0.235	0.286	0.326	0.358	0.386	0.477	0.456	0.317	0.250	0.181	N/A	(40)	
		taud	1.871	2.049	2.073	2.081	2.263	2.348	2.044	2.050	2.465	2.196	1.893	N/A	(41)	
	Ebn at Noon		60	192	239	259	265	260	225	213	237	185	56	0	(42)	
		Edh at Noon	5	17	29	38	36	34	44	40	21	14	4	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		24	247	821	1541	1804	1833	1682	1145	669	222	57	0	(44)	
	RadStd		2	14	36	63	133	158	130	140	66	36	4	0	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page